Problem Statement: Write a program to solve a fractional Knapsack problem using a greedy method.

Code:

```
import java.util.*;

class Item {
    int value;
    int weight;
    double ratio;

    Item(int value, int weight) {
        this.value = value;
        this.weight = weight;
        this.ratio = (double) value / weight;
    }
}

public class FractionalKnapsack {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        double capacity = 60;

        System.out.print("Enter number of items: ");
        int n = sc.nextInt();

        Item[] items = new Item[n];


        for (int i = 0; i < n; i++) {
            System.out.print("Enter value of item " + (i + 1) + ": ");
            int value = sc.nextInt();

            System.out.print("Enter weight of item " + (i + 1) + ": ");
            int weight = sc.nextInt();

            items[i] = new Item(value, weight);
        }


        Arrays.sort(items, (a, b) -> Double.compare(b.ratio, a.ratio));

        System.out.println("\nItems after sorting by value/weight ratio:");
        System.out.println("Item\tValue\tWeight\tRatio");
```

```
        for (int i = 0; i < n; i++) {
            System.out.printf("%d\t%d\t%d\t%.2f\n",
                    i + 1,
                    items[i].value,
                    items[i].weight,
                    items[i].ratio);
        }


        double totalValue = 0;

        for (int i = 0; i < n; i++) {

            if (capacity == 0)
                break;

            if (items[i].weight <= capacity) {


                capacity -= items[i].weight;
                totalValue += items[i].value;

                System.out.println("\nItem " + (i + 1) +
                        " taken completely.");

            } else {


                double fraction = capacity / items[i].weight;

                totalValue += items[i].value * fraction;

                System.out.printf("\n%.2f fraction of Item %d taken.",
                        fraction, i + 1);

                capacity = 0;
            }
        }

        System.out.printf("\n\nMaximum value in knapsack = %.2f\n",
                totalValue);

        sc.close();
    }
}
```

Output:

```
comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ javac FractionalKnapsack.java
comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ java FractionalKnapsack
Enter number of items: 4
Enter value of item 1: 100
Enter weight of item 1: 20
Enter value of item 2: 120
Enter weight of item 2: 30
Enter value of item 3: 150
Enter weight of item 3: 60
Enter value of item 4: 180
Enter weight of item 4: 80

Items after sorting by value/weight ratio:
Item    Value    Weight    Ratio
1    100    20    5.00
2    120    30    4.00
3    150    60    2.50
4    180    80    2.25

Item 1 taken completely.

Item 2 taken completely.

0.17 fraction of Item 3 taken.

Maximum value in knapsack = 245.00
```